

Question 1:

You are given two relational tables, **Authors** and **Books**.

The **Authors** table contains three columns, 'author_id', 'name' and 'country':

author_id	name	country
1	Haruki Murakami	Japan
2	George Orwell	UK

The **Books** table contains three columns, 'book_id', 'title', and 'author_id':

book_id	title	author_id
101	Kafka On the Shore	1
102	1984	2
103	Norwegian Wood	1

How would you best represent this data, taking into consideration both tables, in the JSON format?

The correct answer is:

```
{
  "authors": [
    {
      "author_id": 1,
      "name": "Haruki Murakami",
      "country": "Japan",
      "books": [
        { "book_id": 101, "title": "Kafka on the Shore" },
        { "book_id": 103, "title": "Norwegian Wood" }
      ]
    },
    {
      "author_id": 2,
      "name": "George Orwell",
      "country": "UK",
      "books": [
        { "book_id": 102, "title": "1984" }
      ]
    }
  ]
}
```

Question 2:

You are building a classifier and want to rank features based on how informative they are for predicting the target variable. One of the methods you consider is mutual information. Which of the following statements are true regarding mutual information?

Select one or more:

- a. Mutual information is zero if and only if the variables are statistically independent.

- b. Mutual information ranks features based only on linear relationships, similar to Pearson correlation, which makes it unsuitable for detecting nonlinear dependencies
- c. Mutual information can only be computed for numerical variables and cannot be applied to categorical data.
- d. Mutual information measures both linear and nonlinear dependencies between features and the target, unlike methods such as Pearson correlation, which only capture linear relationships.
- Correct: a) and d)

Question 3:

Which of the following statements are true?

Select one or more:

- a. The K-means algorithm is inherently limited in the shapes of clusters it can identify, as it can only successfully find clusters that are convex.
- b. Soft Clustering assigns a single, absolute cluster label to a data point, while Hard Clustering assigns a probability distribution over all clusters.
- c. Dimensionality Reduction is an unsupervised task where the goal is to compute a function $f(X)$ that results in a continuous, simpler representation y .
- d. Clustering is an unsupervised task used to learn classification y based on labeled training data (X,y)
- Correct: a) and c)

Question 4:

Which of the following statements correctly describe the concepts used to represent clusters or define their "nearness"?

Select one or more:

- a. A cluster can be represented by its Centroid (an *existing data point* closest to others, useful in non-Euclidean spaces) or by a Clustroid (an *artificial point* calculated as the average of all data points).
- b. One way to define "nearness" (Intercluster Distance) of two clusters is by calculating the minimum distance between any two points, one from each cluster (known as single-linkage).
- c. Agglomerative Hierarchical Clustering is highly resistant to the curse of dimensionality because the cluster distance calculation is based only on a select few cluster representatives (centroids/clustroids).
- d. The Agglomerative approach begins with all data points as individual clusters and iteratively merges them, while the Divisive approach starts with one cluster and recursively splits it.
- Correct: b) and d)

Question 5:

- The correct answer is: It uses $\geq p$ instead of $\leq p$, so it keeps the very frequent words instead of removing them.

Question 6:

You are working with a text dataset that contains three sentences:

1. The weather today is sunny!
2. It rained yesterday.
3. I am going shopping today.

You want to analyze the similarity between the documents so you perform simple preprocessing consisting of punctuation removal, lowercasing all letters and splitting by space to obtain words. Then, you construct the TF-IDF matrix and use document features to compute the cosine similarity between the documents. Select the correct statement(s).

Select one or more:

- a. Cosine similarity between (1) and (3) is equal to zero
- b. Cosine similarity between (2) and (3) is equal to zero
- c. Cosine similarity between (1) and (2) is non-zero
- d. Cosine similarity between (1) and (2) is equal to zero
- Correct: b) and d)

Question 7:

You compute rdd2 once and use it for multiple actions:

```
rdd2 = rdd1.map(f1)
list1 = rdd2.filter(f2).collect()
list2 = rdd2.filter(f3).collect()
```

What is the best fix to avoid recomputing map(f1) multiple times?

Select one:

- a. Call rdd1.persist() instead of rdd2.persist()
- b. Add rdd2.persist() after the map
- c. Replace both filters with one filter
- d. Replace collect() with count()
- Correct: b)

Question 8:

Consider the following operation on an RDD:

```
map(f), filter(g), reduce(h), count(), take(10)
```

Select one:

- a. Transformations: map, filter, take — Actions: reduce, count, take
- b. Transformations: map, filter — Actions: reduce, count, take
- c. Transformations: map, filter, reduce — Actions: count, take
- d. Transformations: map, filter, count — Actions: reduce, take
- Correct: Transformations: map, filter — Actions: reduce, count, take

Question 9:

In a Spark application, you are tasked with counting the total number of log entries containing the string "ERROR" across a large, distributed RDD named log_rdd. You write the following two PySpark code snippets to attempt this task:

Snippet 1 (Using Standard Variable):

```
error_count_1 = 0

def count_errors_1(line):
    # This closure attempts to update the global variable
    if "ERROR" in line:
        error_count_1 += 1
```

```
log_rdd.foreach(count_errors_1)
print(error_count_1)
```

Snippet 2 (Using Spark Accumulator):

```
sc is the SparkContext
error_count_2 = sc.accumulator(0)

def count_errors_2(line):
    # This closure uses the Spark-provided tool for aggregation
    if "ERROR" in line:
        error_count_2.add(1)

log_rdd.foreach(count_errors_2)
print(error_count_2.value)
```

Which of the following statements correctly describe the final result obtained from these snippets in a distributed Spark cluster environment? Correct:

- b. Snippet 2 will always print the correct total number of "ERROR" entries.
- c. Snippet 1 will typically print zero (0) or an incomplete count because the error_count_1 variable is copied to each worker process, and the updates made on the workers are not communicated back to the driver program.
- d. Snippet 2 correctly uses an Accumulator, which is designed specifically as a write-only, distributable variable for efficiently aggregating values (like counters and sums) from the worker processes back to the driver.

Question 10:

Below is a simplified cluster-centroid table (z-scored units) for the 4 post-mortem curve features:

Cluster	Premortem Mean	Short-Term Boost	Long-Term Boost	Halving Time
C1	-0.2	+1.8	-0.1	-1.2
C2	-1.1	-0.4	-0.7	+0.1
C3	+0.3	+0.9	+1.7	+0.9
C4	+1.9	-0.8	-0.6	+0.2

Which interpretation is most accurate?

- The correct answer is: C3 represents "rise" curves: moderate short-term boost and unusually high long-term boost

Question 11:

A linear classifier for sentiment has weight vector β (in $\mathbb{R}^3$) for three words {"great", "awful", "movie"}. You observe the following in your training set:

Word	Appears only in documents with sentiment:
great	5
awful	1
movie	1-5

The learned weights without regularization are:

$$\beta = (\beta_1 = 12.0, \beta_2 = -10.0, \beta_3 = 0.1)$$

You now add L2 regularization with $\lambda = 0.5$ to the loss:

$$L(\beta) = \sum_i (y_i - x_i^T \beta)^2 + \lambda \|\beta\|_2^2$$

Question: Which of the following outcomes is **impossible** after retraining? (Multiple answers)

Select one:

- a. The training error increases
- b. β_3 becomes exactly zero
- c. β_2 decreases in magnitude
- d. The test error decreases
- e. β_1 decreases in magnitude
- Answer: L2 regularization (ridge) can shrink weights, but it cannot make a non-zero weight become exactly zero.
-> β_3 becomes exactly zero

Question 12:

A researcher is plotting the distribution of city populations, which follows a heavy-tailed power law. She decides to use log-log axes. Why is this transformation helpful?

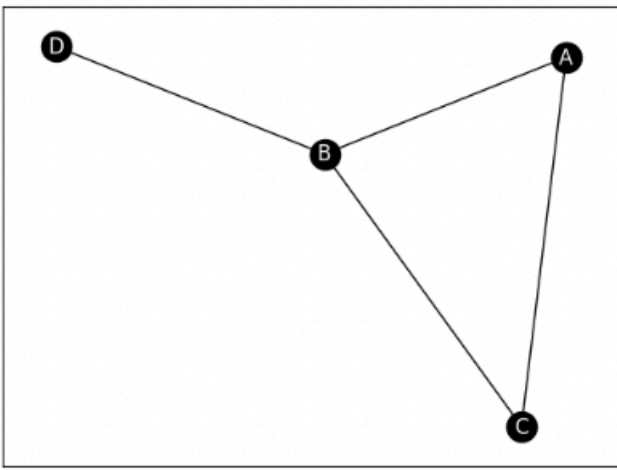
Select one:

- a. It normalizes the data so all values are between 0 and 1
- b. It removes outliers automatically from the dataset
- c. It makes the distribution appear bell-shaped and easier to interpret
- d. It linearizes the power-law relationship, making the slope represent the exponent
- Correct: d)

Question 13:

You work for a logistics company and you are tasked with analyzing their distribution network. You are given a graph containing nodes (cities) A, B, C and D and edges (roads) which connect them and you want to analyze the load of the traffic through the cities in the network. You know that the company always uses the shortest path between two cities when planning the transfer of products so you opt to use the betweenness centrality. Given the graph in the picture, what is the betweenness centrality of the nodes in the network (answers follow order [node A, node B, node

C, node D]])?



Select one:

- ☒ a. [2/3, 1, 2/3, 1/6] ❌
- ☐ b. [1/3, 2/3, 1/3, 0]
- ☐ c. [0, 2/3, 0, 0]
- ☐ d. [0, 3, 0, 0]

$$BC(v) = \sum_{\substack{s \neq v \neq t \\ s \neq t}} \frac{\# \text{ shortest paths from } s \text{ to } t \text{ that go through } v}{\# \text{ shortest paths from } s \text{ to } t}$$

- The correct answer is: [0, 2/3, 0, 0]

Question 14:

We use a logistic regression to model the probability p of a disease being present. The result is: $\text{logit}(p) = -3 + 0.02 \times X_1 + 0.5 \times X_2$

where X_1 is the person's age in years, and X_2 indicates if the person is a smoker ($X_2 = 1$) or not ($X_2 = 0$). Which of the following statements is true?

Select one:

- a. The odds of having the disease for smokers is approximately three times that for non-smokers.
- b. Aging by one year increases the odds by approximately 2%.
- c. The log-odds increase by 50% for smokers.
- d. Being a non-smoker increases the log-odds by 0.5.

- The correct answer is: Aging by one year increases the odds by approximately 2%.

$$\log(p/(1-p)) = -3 + 0.02X_1 + 0.5X_2$$

In logistic regression: A coefficient β means the **odds** are multiplied by e^β for a 1-unit increase in that variable.

- Effect of age (X_1)
 - $e^{0.02} \approx 1.0202$
 - So **each additional year multiplies the odds by ≈ 1.02** , i.e., about a **2% increase**.
- Effect of smoking (X_2)
 - $e^{0.5} \approx 1.65$
 - So smokers have **1.65× the odds** of non-smokers (65% higher odds).
- c. "The log-odds increase by 50% for smokers."** False ❌
 - Smoking adds **+0.5** to the log-odds, not **+50%**.

Question 15:

In propensity score matching, two subjects with equal propensity scores will have:

Select one:

- a. Identical covariates
- b. Similar distribution of observed covariates
- c. Identical treatment effects
- d. Equal outcome
- The correct answer is: Similar distribution of observed covariates

Question 16:

Which of the following are valid examples of confounders in an observational study?

Select one:

- a. A variable that affects both the probability of receiving the treatment and the outcome
- b. A variable that affects treatment assignment but has no effect on the outcome
- c. A variable that affects the outcome but is completely unrelated to treatment assignment
- d. A variable measured after the treatment is applied
- The correct answer is: A variable that affects both the probability of receiving the treatment and the outcome

Question 17:

Which of the following statements about logistic regression are correct?

Select one:

- a. Logistic regression assumes that each class's points are generated from a Gaussian distribution.
- b. Models the log-odds as a non-linear function.
- c. The cost function of logistic regression is concave.
- d. Logistic regression is equivalent to a neural network without hidden units and using cross-entropy loss
- The correct answer is: Logistic regression is equivalent to a neural network without hidden units and using cross-entropy loss.

Question 18:

When is accuracy NOT an appropriate evaluation metric? (Select all that apply)

Select one or more:

- a. When classes are highly imbalanced/skewed
- b. When detecting rare events like fraud
- c. When false positives and false negatives have different costs
- d. When you have a binary classification problem
- e. When all errors are equally important and classes are balanced
- The correct answers are: a), b) and c)

Question 19:

Which of the following statement(s) are true regarding cross-validation?

Select one or more:

- a. It is guaranteed to prevent overfitting
- b. Reporting performance from the same cross-validation used to tune hyperparameters yields an unbiased estimate of generalization error
- c. It is often used to select hyperparameters
- d. It is useful when there is not enough data for a train, validation, and test split
- Correct: c) and d)

Question 20:

You deploy a medical classifier for a rare disease (1% positives). On a test set of 1,000 patients, your model outputs:

- Predicted positive: 50
 - True positive (TP): 5
 - False positive (FP): 45
- Predicted negative: 950
 - False negative (FN): 5
 - True negative (TN): 945

Which statement is more accurate?: Accuracy is high, but precision is low; the model is risky if false alarms are costly